

Neck MR Angiography (MRA)

ACG: A-0034 (AC)

[Link to Codes](#)

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Clinical Indications

- Neck MR angiography (MRA) may be indicated for **1 or more** of the following(1)(2):
 - ☐ Carotid artery stenosis or occlusion, suspected, as indicated by **ALL** of the following(5)(23)(24)(25)(26)(27):[N](#)
 - MRA results will impact patient management.(37)(38)
 - Signs and symptoms of carotid artery stenosis or occlusion, as indicated by **1 or more** of the following(39)(40)(41):
 - Cognitive impairment
 - Confusion
 - Difficulty speaking or understanding speech(38)
 - Dizziness
 - Gait disturbance(38)
 - Headache that is sudden, severe, and unexplained
 - Loss of balance or coordination
 - Loss of consciousness
 - Numbness, weakness, or paralysis of face, arm, or leg, unilateral(38)
 - Visual disturbance, particularly unilateral(38)
 - Carotid or vertebral arterial aneurysm, suspected, or follow-up of known disease(4)(42)[N](#)
 - Carotid or vertebral arterial dissection, suspected, or follow-up of known disease(27)(43)(44)(45)[N](#)
 - Carotid or vertebral thromboembolism, suspected, or follow-up of known disease(50)[N](#)
 - Carotid or vertebral vasculitis or collagen vascular disease, suspected(51)(52)(53)(54)(55)[N](#)
 - Congenital anomalies of carotid and vertebrobasilar circulations, suspected, or follow-up of known disease(58)[N](#)
 - Internal jugular vein thrombosis or compression, suspected, or follow-up of known disease[N](#)
 - Intramural hematoma of carotid or vertebrobasilar circulation, suspected, or follow-up of known disease(60)(61)[N](#)
 - ☐ Neck trauma evaluation needed, as indicated by **1 or more** of the following(62):[N](#)
 - Blunt neck injury, with suspicion of carotid or vertebral artery injury(63)
 - Initial screening study in patient age 18 years or younger with suspected carotid artery injury(64)
 - Penetrating neck injury(65)
 - ☐ Postoperative evaluation following carotid endarterectomy, as indicated by **1 or more** of the following(23):[N](#)
 - Need to assess arterial patency and rule out development of new lesions; intervals include 1 month postoperatively, 6 months postoperatively, and annually thereafter
 - New signs or symptoms of carotid artery stenosis
 - Substitute for otherwise indicated catheter angiography

- ☐ Preoperative or preprocedural planning for carotid endarterectomy or percutaneous intervention needed, as indicated by **1 or more** of the following(23)(66)(67):[N](#)
 - Acute stroke, and performed in conjunction with head MRA
 - Duplex (Doppler) scan finding of high-grade stenosis in asymptomatic patient
 - Duplex (Doppler) scan findings indeterminate
 - Duplex (Doppler) scan or clinical findings show high carotid bifurcation, loops, or kinks.
 - History of neck radiation therapy
- Vascular neck mass, suspected (eg, hemangioma, lymphangioma, arteriovenous malformation, carotid body tumor)(58)(69)[N](#)
- ☐ Vertebral artery stenosis or occlusion, suspected, as indicated by **ALL** of the following(23):[N](#)
 - MRA results will impact patient management.
 - Signs or symptoms of vertebral artery stenosis or occlusion, as indicated by **1 or more** of the following:
 - Ataxia
 - Diplopia
 - Dysarthria
 - Dysphagia
 - Facial numbness or paresthesia
 - Hearing loss (acute), sensorineural
 - Limb and trunk sensory deficits
 - Loss of taste
 - Motor paresis
 - Nystagmus
 - Syncope
 - Vertigo(70)
 - Visual field defects
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

MRA uses MRI technology to produce detailed 2-dimensional or 3-dimensional images of the vascular system and perivascular anatomy without the risks usually associated with catheter angiography, including radiation exposure and arterial puncture.(3)(4) **(EG 2)** The gadolinium-based contrast material used for MRA carries a lower risk of allergic reaction than the iodine-based contrast material used in catheter angiography and CTA.(5)(6) **(EG 2)** Some gadolinium-based contrast agents used for diagnostic and treatment guidance with MRI or MRA have been associated with nephrogenic systemic fibrosis in patients with chronic kidney disease or acute kidney injury.(6)(7)(8)(9)(10) **(EG 2)** Gadolinium deposition in the brain has been reported by multiple research groups; however, the clinical significance of this finding is unclear.(9)(10)(11)(12) **(EG 2)** Specialty society guidelines recommend weighing the risks vs benefits when using gadolinium-based contrast agents for MRI.(13) **(EG 2)** MRI-based techniques may be contraindicated in patients with ferromagnetic intracranial aneurysm clips and certain types of ferromagnetic foreign bodies and cochlear implants.(14) **(EG 2)** Although these implants have traditionally included cardiac pacemakers and implantable cardioverter-defibrillators, multiple trials have validated the safe and effective use of pacemaker systems designed for MRI use, and a guideline supported by multiple specialty societies states that many patients with implantable cardiac devices, including those not originally intended for MRI scanning, can safely undergo MRI scanning under certain clinical situations.(15)(16)(17)(18)(19) **(EG 1)** Time-of-flight MRA, which does not require the injection of a contrast medium, and other noncontrast techniques are alternatives for patients with contraindications to receiving contrast.(4)(5)(20) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 53 published peer reviewed articles, 11 specialty society or other evidence-based guidelines, and 1 book section.

For carotid or vertebral artery stenosis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Noninvasive imaging has largely replaced traditional digital subtraction angiography due to decreased risk of complications.(23)(25)(28) **(EG 2)** Both MRA and CTA are sensitive and specific noninvasive imaging modalities for the assessment of the location and degree of carotid or vertebral artery stenosis and are used if Doppler studies are indeterminate.(29)(30)(31) **(EG 2)** A systematic review and meta-analysis of studies assessing the diagnostic accuracy of MRA for internal carotid artery disease found overall sensitivity of 91% to 95% and specificity of 88% to 92% for stenoses of 70% to 99%; in addition, the review found sensitivity of 95% to 99% and specificity of 99%

to 100% for total occlusion. Compared with contrast-enhanced MRA, time-of-flight MRA has a lower sensitivity and specificity for moderate-grade and severe-grade stenosis as well as total occlusion.(32) **(EG 1)** A review article notes that gadolinium-enhanced MRA is highly accurate in identifying carotid stenosis amenable to surgical intervention, but is inferior to CTA for detection of intraluminal thrombus and occlusion.(33) **(EG 2)** For vertebral artery stenosis, a prospective cohort study comparing various noninvasive diagnostic tests showed that MRA was more sensitive and specific than either CTA or duplex ultrasonography for detection of stenoses.(34) **(EG 2)** A review article notes that CTA and MRA of the head and neck are useful imaging modalities for diagnosis and management of acute ischemic stroke.(28) **(EG 2)** Practice guidelines recommend that MRA or CTA of the intracranial and extracranial vessels (eg, the carotid arteries) be available at primary stroke centers for admitted stroke patients who may benefit from such testing when needed to guide or determine treatment decisions.(35)(36) **(EG 2)** A specialty society guideline notes that MRA of the head and neck is appropriate in the evaluation of acute stroke or transient ischemic attack and states that combined cerebral and vascular evaluation should be considered.(27) **(EG 2)**

For carotid or vertebral arterial aneurysms, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles note that MRA is useful for the evaluation of cervical and intracranial vasculature due to its ability to provide excellent vascular contrast and detail without the use of ionizing radiation.(4)(42) **(EG 2)** A specialty society guideline states that neck MRA is appropriate for the evaluation of extracranial aneurysms and vascular malformations.(1) **(EG 2)**

For carotid or vertebral arterial dissection, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRA and multidetector CTA have emerged as less invasive, more rapid, and possibly more revealing means of assessing disease.(46) (47)(48) **(EG 2)** A review article notes that MRA has demonstrated sensitivity up to 95% and specificity up to 99% for assessing luminal changes related to arterial dissection.(49) **(EG 2)** A specialty society guideline notes that MRA of the neck is appropriate in the evaluation of suspected cervical vascular dissection or injury as well as ongoing surveillance of these cases owing to its high tissue contrast, which may be helpful in the delineation of subtle intramural hematomas that may not be visible in other studies.(27) **(EG 2)**

For carotid or vertebral thromboembolism, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that MRA is appropriate for the evaluation of the presence and extent of thromboembolic phenomena in the cervicocerebral vasculature.(1) **(EG 2)**

For carotid or vertebral vasculitis or collagen vascular disease, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that head and neck MRA are appropriate for the evaluation of vasculitis and collagen vascular diseases, particularly in pediatric patients.(1) **(EG 2)** A review article notes that MRA can be useful in detecting fibromuscular dysplasia of the carotid arteries.(56) **(EG 2)** Review articles note that contrast-enhanced MRA can provide information regarding anatomic variations, stenoses, and occlusions in patients with supra-aortic vasculitis, including Takayasu and giant cell arteritis.(4)(49)(57) **(EG 2)**

For congenital anomalies of the carotid and vertebrobasilar circulations, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A review article notes that the development of 4-dimensional MRA, which refers to the acquisition of volume data sets at high image refresh rates, has allowed for simultaneous anatomic and functional analysis of several vascular anomalies, including arteriovenous malformations.(59) **(EG 2)** A specialty society guideline states that MRA is appropriate for the evaluation of congenital vascular anomalies and vascular anatomic variants.(1) **(EG 2)**

For internal jugular vein thrombosis or compression, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that MRA is appropriate for the evaluation of traumatic injuries to cervical arteries and veins, such as those that may result in thrombosis or compression of the jugular veins.(1) **(EG 2)**

For intramural hematoma of the carotid or vertebrobasilar circulation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** MRA can identify an intramural hematoma of the carotid or vertebrobasilar circulation, such as what may occur following vascular dissections.(60) **(EG 2)** A specialty society guideline states that MRA is appropriate for the evaluation of traumatic injuries to cervicocerebral vessels, including dissections.(1) **(EG 2)**

For neck trauma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** While digital subtraction angiography remains the gold standard for evaluation of cervical vessels, MRA is one of several appropriate initial imaging modalities and is commonly used in the emergency setting for evaluation of blunt or penetrating injury.(62)(63) **(EG 2)** A specialty society guideline states that MRA is appropriate for the evaluation of traumatic injuries to cervicocerebral vessels, including dissections.(1) **(EG 2)**

For postoperative evaluation following carotid endarterectomy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline states that it is reasonable to obtain noninvasive imaging, such as MRA, of the extracranial carotid arteries following carotid endarterectomy at intervals of 1 month, 6 months, and annually thereafter to assess arterial patency and rule out the development of new lesions.(23) **(EG 2)**

For preoperative or preprocedural planning for carotid endarterectomy or percutaneous intervention, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline notes that, when intervention is planned for significant carotid stenosis detected by duplex sonography, CTA, MRA, or digital subtraction angiography can be useful to evaluate stenosis severity and to identify intracranial or intrathoracic vascular lesions that are inadequately assessed by duplex sonography.(23) **(EG 2)** A review article notes that MRA is a viable option for preoperative cross-sectional imaging of the carotid arteries.(68) **(EG 2)** A specialty society guideline states that knowledge of vessel anatomy and the presence of extracranial vessel dissections, stenoses, and occlusions can assist in planning endovascular procedures.(67) **(EG 2)**

For vascular neck masses, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A review article notes that MRA, for which several different image-enhancement techniques are available, is an appropriate imaging modality for the evaluation of neck masses.(69) **(EG 2)** A specialty society guideline states that MRA is appropriate for the evaluation of neck masses in pediatric and adult patients, particularly to define vascular supply.(1) **(EG 2)**

For vertebral artery disease, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society guideline recommends that MRA or CTA imaging be performed rather than ultrasound imaging for evaluation of the vertebral arteries in patients with suspected posterior cerebral or cerebellar ischemia.(23) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For carotid artery plaque morphology, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** Specialty society guidelines note that MRA is unable to define calcium within a carotid plaque, and although MRA can be used to analyze other elements of carotid plaque morphology, selection of therapy as determined by morphology has not been shown to improve clinical outcomes.(5)(21) **(EG 2)** A systematic review of 19 studies (2601 patients) evaluating the efficacy of MRA for the characterization of symptomatic carotid artery plaques found that MRA with vessel wall imaging sequences can identify plaque features that may serve as markers for increased risk for stroke but that further studies are needed to determine the usefulness of these markers as predictors of neurovascular events among asymptomatic patients.(22) **(EG 1)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(71); 42 CFR 422.101(72)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(73); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(74); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(75)

Medicare Coverage Determinations: Medicare Coverage Database(76)

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